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Analyzing Human Motion for Predicting Penalty Direction in Football

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8 credits

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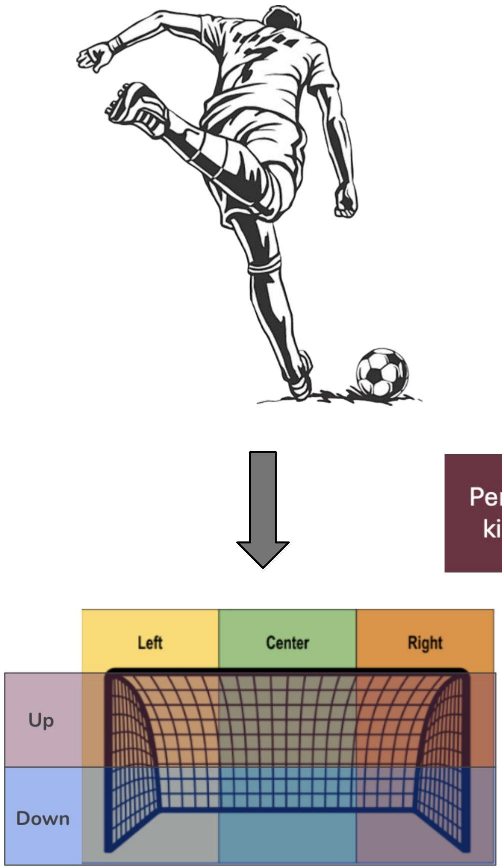
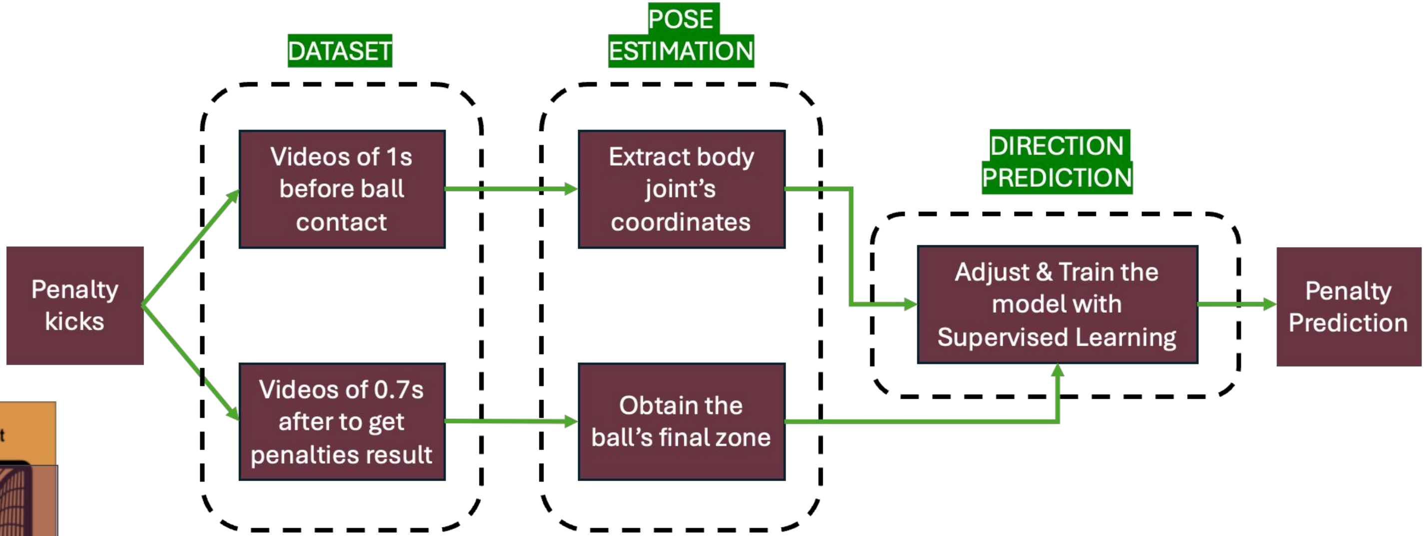
8 credits

What is the problem

Predicting the direction of a penalty kick solely based on the kicker's run-up remains a challenging task in sports analytics.



Detailed Statement: "The project aims to predict the direction of a penalty kick by analyzing the pose and run-up of the kicker in the second before the kick."



Quick reminder :

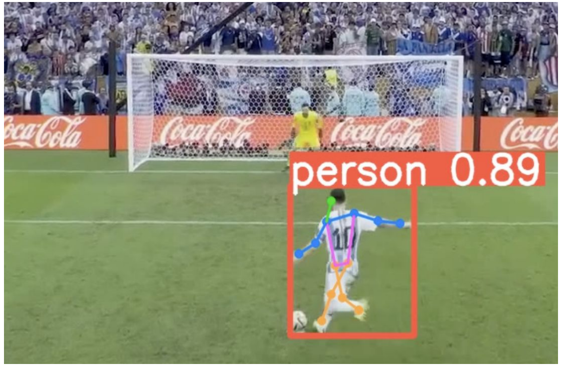
Already accomplished	**What is left**
-> Collecting data : 1000 FIFA videos and some real videos but of poor quality for the pose estimation **->** Standardization of videos **->** Initial development of the pose estimation model	**->** Improve the pose estimation model for real and fifa videos **->** Expand dataset & enhance the automatic labeling **->** Implement & train the model based on ST_Trans

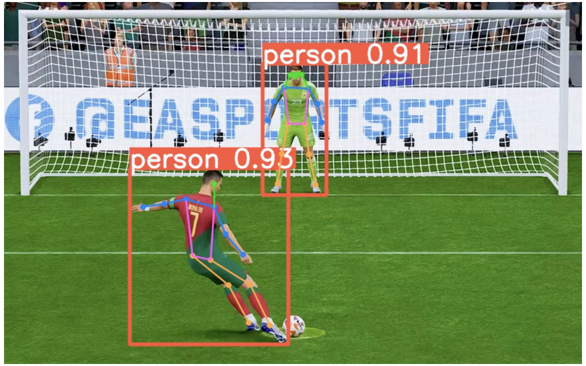
EPFL VITA

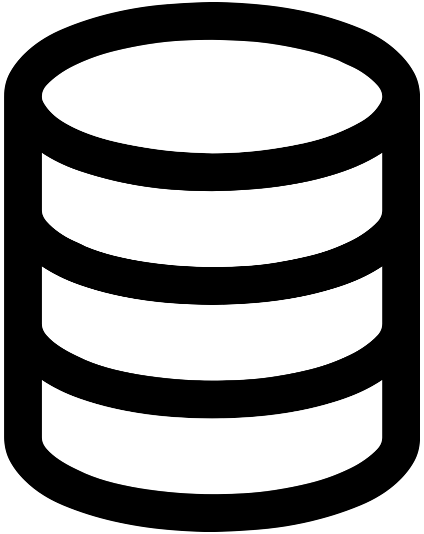
Since the midterm presentation

Pose estimation

- Implementation of a new pose estimation model
- Yolov8 (latest version of pose estimation model)
- Extracting important key points
- Implementing certainty score for each key point



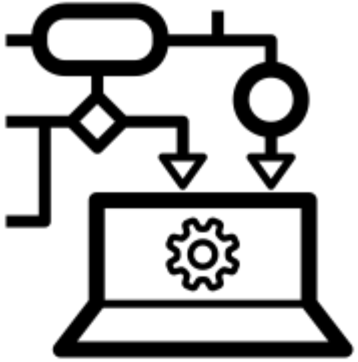




Expanding Dataset

- Standardization of the input :
  - 30 frames : 12 body joints coordinates (x,y, certainty score)
  - 30*36 → 1’080 inputs per penalty
- Enhancement of automatic labeling
- More than 1’500 videos in total :
  - 125 penalties from real footage
  - 1’385 from FIFA 23

Model Implementation



- Inspired from ST_Trans model
- 4 convolutional layers for feature extraction
- 8 transformer layers to focus on different aspects of the data
- 8 residuals blocks to refine learned features across layers
- Unbalanced dataset :
  - To avoid overfitting → reduced to 4 zones, removal of BC & TC.
- Tuning of hyperparameters (learning rate, etc) with Optuna.

37.29 %	2.03 %	17.04 %
21.42 %	0.23 %	21.81 %

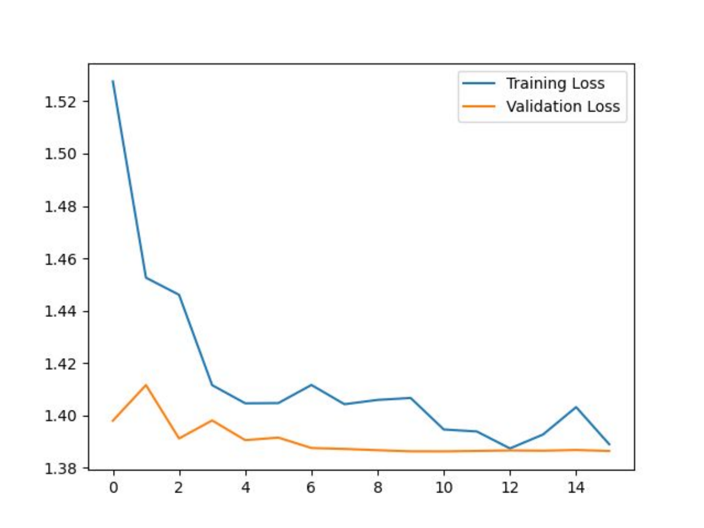
Results

Training data : 1’000 penalties

Validation data : 100 penalties

Testing data : 100 penalties

- Accuracy : 35 %
- Improvement of only 10%



Limitations & Improvements

1. Unbalanced set $\longrightarrow$ data collection
2. Data augmentation : symmetrical inversion, rotation, etc.
3. Dimensionality reduction : PCA, etc.
4. 3D pose estimation instead of 2D
5. Integration of additional features (right-footed or left-footed for instance)